ModelIt!

Teacher's Guide

Can Plants Survive Without Bees?

The Amazing World of Pollination

Quick Reference

Grade Level	Grade 3
NGSS Standard	3-LS2-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 3-LS2-1

Construct an argument that some animals form groups that help members survive.

Three-Dimensional Learning

Science Practice	**Engaging in Argument from Evidence** — Students construct arguments using model evidence that pollinators are essential for plant reproduction and food production, supporting their claims with simulation data.
Disciplinary Core Idea	**LS2.A Interdependent Relationships in Ecosystems** — The food of almost any kind of animal can be traced back to plants. Organisms are related in food webs in which some animals eat plants for food and other animals eat the animals that eat plants. Many organisms can survive only in their particular habitats.
Crosscutting Concept	**Systems and System Models** — Students model the pollination system showing how pollinators, flowers, and fruit production are interconnected components that depend on each other to function.

CAST Testing Alignment

MULTIPLE CHOICE:

A farmer notices that one apple tree near a wildflower meadow produces lots of apples, but an identical tree far from flowers produces very few. Which best explains the difference?

CONSTRUCTED RESPONSE:

Explain what would happen to a strawberry farm if all the bees in the area disappeared. Use the words 'pollinator,' 'pollination,' and 'fruit production' in your answer.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Pollination is one of nature's most important partnerships. Without pollinators, most of the foods we eat and flowers we enjoy would simply not exist. Understanding this relationship helps us protect both plants and pollinators.

What Is Pollination?

Pollination is the transfer of pollen from the male part of a flower (anther) to the female part (stigma), which allows the plant to make seeds and fruit. While some plants can pollinate themselves or use wind, about 87.5% of flowering plants need animal pollinators. Bees are the most important pollinators, but butterflies, moths, hummingbirds, bats, beetles, and even ants also pollinate plants. When a bee lands on a flower to drink nectar, pollen sticks to its fuzzy body and transfers to the next flower it visits.

Why Pollinators Matter for Food

About 75% of the world's food crops depend at least partly on pollination. Without bees and other pollinators, we would lose apples, almonds, blueberries, strawberries, chocolate, coffee, and many more foods. One out of every three bites of food you eat exists because a pollinator helped make it. The economic value of pollination to agriculture is estimated at over $200 billion worldwide each year.

Threats to Pollinators

Pollinator populations are declining worldwide due to habitat loss, pesticide use, disease, and climate change. Bees are losing wildflower meadows where they feed. Pesticides called neonicotinoids can harm bees' navigation and memory. A disease called colony collapse disorder has killed millions of honeybee colonies. Scientists and farmers are working to protect pollinators by reducing pesticide use, planting pollinator gardens, and creating bee-friendly habitats.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify pollinator visits as an external component and pollination success and fruit production as internal components that change based on how many pollinators visit the flowers.

E **Establish Relationships**
Students determine that pollinator visits positively affect pollination success, and pollination success positively affects fruit production — more bee visits means more flowers get pollinated means more fruit.

V **Visualize & Model**
Students build a model showing how increasing pollinator visits leads to greater pollination success and higher fruit production.

E **Evaluate Outcomes**
Students run no pollinators, some pollinators, and many pollinators scenarios to observe how the number of visits affects pollination and fruit across the system.

R **Revise & Extend**
Students add flower diversity or pesticide use to explore how garden variety attracts more pollinators while chemicals can reduce their numbers.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Who likes apples? Strawberries? Chocolate? What if I told you that NONE of those would exist without bees?"*

Watch For

Hold up an apple or show a photo. Ask: What does this apple have to do with a bee? Collect ideas.

Slide 2: What Will We Learn Today?

3 min

Say: *"Today we are going to discover why bees are some of the most important animals on the entire planet!"*

Watch For

Read objectives together. Show a close-up image of pollen on a bee's body and explain what pollen is.

Slide 3: The Big Question

3 min

Say: *"If all the bees disappeared tomorrow, would your favorite foods still exist? Let us investigate!"*

Watch For

Show a picture of a grocery store. Ask students to guess which foods need bees. Reveal: about 1 in 3 items would disappear!

Slide 4: Let's Build a Model!

3 min

Say: *"We are going to build a model to test exactly how important pollinators are for making food."*

Watch For

Preview each LEVER step. Explain that scientists study pollination to help protect our food supply.

Slide 5: Activity 1: Sort the Parts

7 min

Say: *"What parts of the pollination system can we identify? What starts the whole process?"*

Watch For

Guide sorting of three components. Explain pollinator visits are external because the number of bees depends on factors outside the plant.

Slide 6: Activity 2: Connect with Arrows

7 min

Say: *"When MORE bees visit, does pollination go up or down? And what about the amount of fruit?"*

Watch For

Help students draw positive arrows from pollinator visits to pollination success, and from pollination to fruit production.

Slide 7: Activity 3: Run the Simulation!

8 min

Say: *"Let us test it! What happens with zero bees, some bees, and LOTS of bees? Predict first!"*

Watch For

Have students predict, then run each scenario. Discuss the dramatic difference between zero and many pollinators.

Slide 8: What Did We Discover?

7 min

Say: *"Now we know — bees are not just making honey, they are making our FOOD! We need to protect them!"*

Watch For

Summarize discoveries. Show the list of foods that depend on bees. Brainstorm ways the class can help pollinators.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Pollinator Visits
- **INTERNAL:** Pollination Success, Fruit Production

Relationships

- **Pollinator Visits to Pollination Success:** POSITIVE (+) — Each time a bee visits a flower, it has a chance to transfer pollen. More visits mean more flowers receive pollen and more pollination events are successful. A flower visited by 10 bees has a much better chance of being pollinated than one visited by only 1.
- **Pollination Success to Fruit Production:** POSITIVE (+) — Only pollinated flowers can produce fruit and seeds. The more flowers that are successfully pollinated, the more fruit the plant produces. An apple tree with 100% pollination success produces a full crop, while one with only 10% pollination produces very few apples.

Key Discoveries

- More pollinator visits lead to more successful pollination, which produces more fruit and seeds
- Without pollinators, many plants cannot reproduce — no bees means no apples, strawberries, or almonds
- Bees and plants are interdependent — bees need flower nectar for food, and flowers need bees to spread pollen
- About 75% of the food crops humans eat depend on pollinators like bees, butterflies, and hummingbirds

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Bees only make honey — that is their purpose"	Honey production is only a small part of what bees do. Their most important role is pollination — transferring pollen between flowers so plants can make fruit and seeds. Without bee pollination, most of our food crops would fail. Honey is a bonus, but pollination is the main reason bees are so essential.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Plants can make fruit without any help from animals"	While some plants use wind or self-pollination, about 87.5% of flowering plants need animal pollinators to reproduce. Fruit only forms after successful pollination. Without pollinators, apple trees would bloom but produce no apples, strawberry plants would flower but grow no berries.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Only honeybees are important pollinators"	There are over 20,000 species of bees worldwide, and many other animals pollinate too. Bumblebees, sweat bees, and mason bees are excellent pollinators. Butterflies, moths, hummingbirds, bats, beetles, and even some lizards pollinate flowers. A healthy ecosystem has MANY types of pollinators working together.